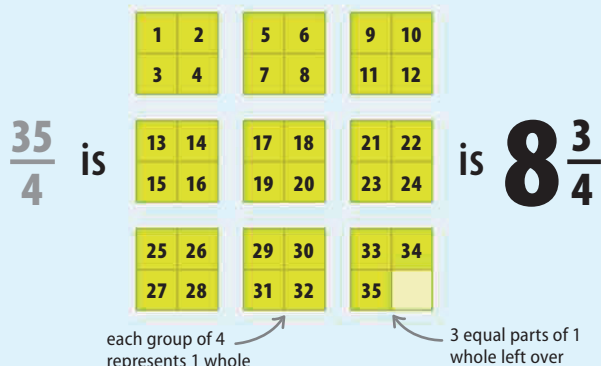


Turning top-heavy fractions into mixed fractions

A top-heavy fraction can be turned into a mixed fraction by dividing the numerator by the denominator.



Draw groups of four numbers—each group represents a whole number. The fraction is eight whole numbers with $\frac{3}{4}$ (three quarters) left over.

numerator

whole number of 8 is produced with 3 left over

$$\frac{35}{4} = 35 \div 4 = 8 \text{ r } 3 = 8\frac{3}{4}$$

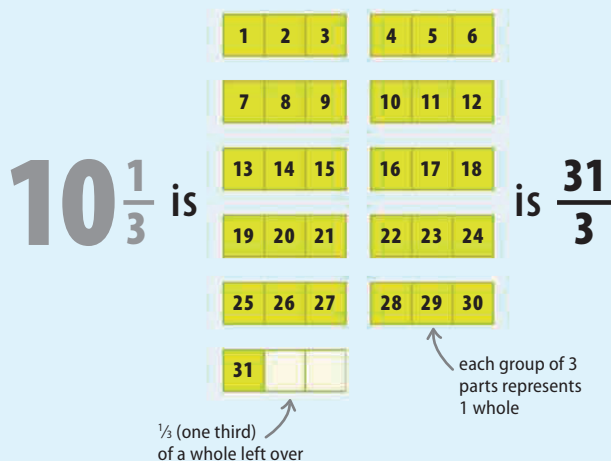
denominator

Divide the numerator by the denominator, in this case, 35 by 4.

The result is the mixed fraction $8\frac{3}{4}$ made up of the whole number 8 and 3 parts—or $\frac{3}{4}$ (three quarters)—left over.

Turning mixed fractions into top-heavy fractions

A mixed fraction can be changed into a top-heavy fraction by multiplying the whole number by the denominator and adding the result to the numerator.



Draw the fraction as ten groups of three parts with one part left over. In this way it is possible to count 31 parts in the fraction.

whole number

multiply whole number by denominator

add to numerator

$$10\frac{1}{3} = \frac{10 \times 3 + 1}{3} = \frac{31}{3}$$

denominator

Multiply the whole number by the denominator—in this case, $10 \times 3 = 30$. Then add the numerator.

The result is the top-heavy fraction $\frac{31}{3}$, with a numerator (31) greater than the denominator (3).